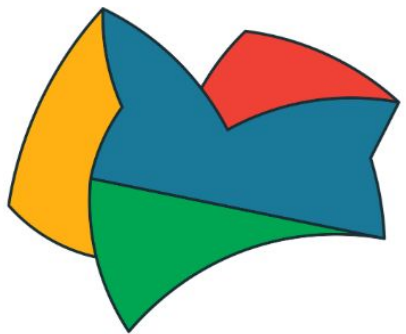




XIII CBTermo

IX Escola de Termodinâmica

**Minicurso: Cálculos Ab Initio para
Termodinâmica**

Aula 01 Introdução ao DFT quântico

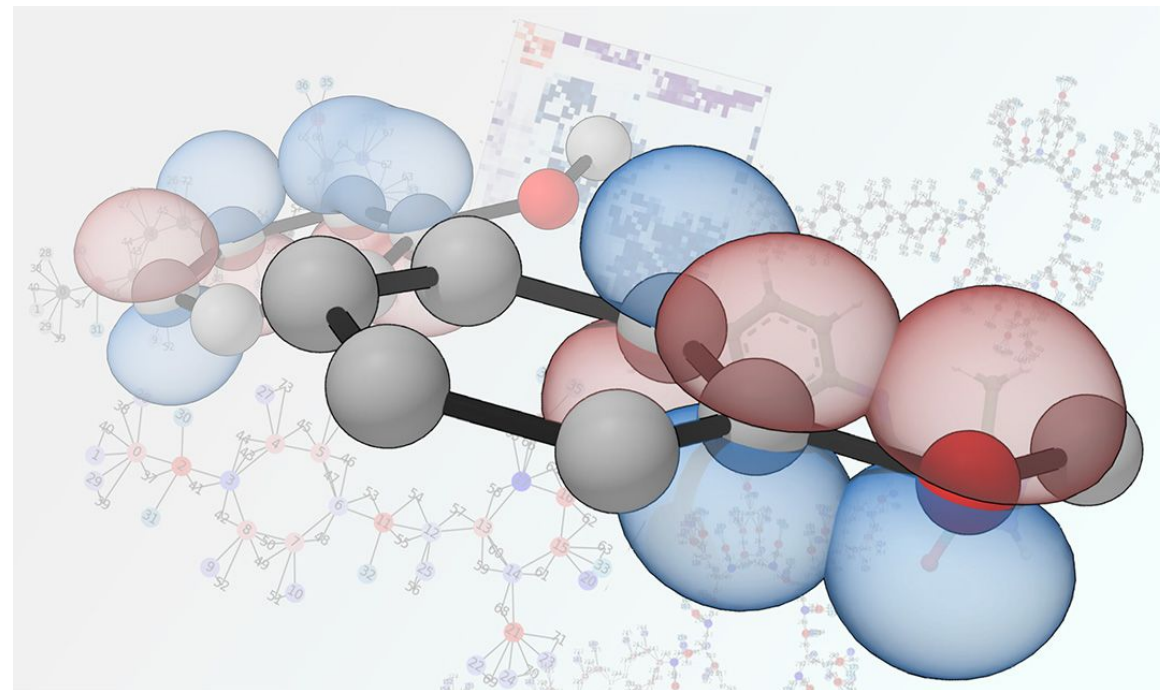
Prof. Elvis Soares

elvis@peq.coppe.ufrj.br

Rio de Janeiro, 21 de maio de 2026

PEQ.

COPPE
UFRJ



$$i\hbar \frac{\partial}{\partial t} |\Psi\rangle = \hat{H} |\Psi\rangle$$

$$|\Psi\rangle = |\psi\rangle e^{-iEt/\hbar}$$

$$\hat{H} |\psi\rangle = E |\psi\rangle$$

De onde eu venho



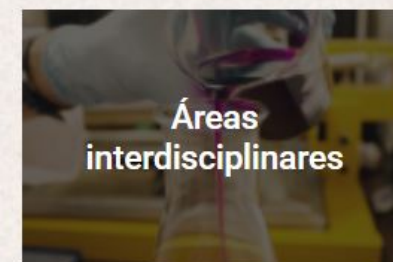
De onde eu venho



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PROGRAMAS DA COPPE





PEQ ▾

Destaques ▾

Admissão ▾

Pesquisa ▾

Informações Acadêmicas ▾

Produção Científica ▾

Escola Piloto



**FALTAM POUCOS DIAS!
AS INSCRIÇÕES PARA O DOUTORADO
PEQ-COPPE/UFRJ 2025/04 ESTÃO
CHEGANDO AO FIM**

CORRA E INSCREVA-SE HOJE MESMO!

Áreas de Pesquisa ▾

Laboratórios Parceiros

Cinética e Catálise

Engenharia de Polímeros

Fenômenos Interfaciais

Modelagem, Simulação e Controle de Processos

Processos Biotecnológicos e Tecnologia Ambiental

Processos de Separação com Membranas e Polímeros

Termodinâmica Aplicada

Termofluidodinâmica



7 professors

10 postdocs

~30 PhD students

~10 MsC students

~20 ICs students



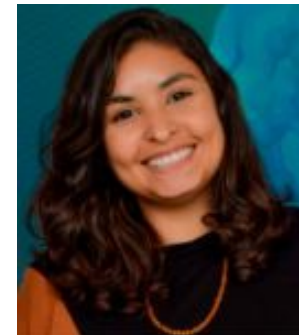
Prof. Fred Tavares



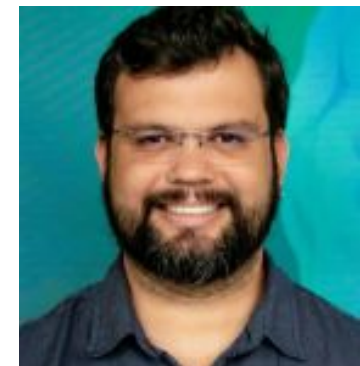
Prof. Amaro Barreto Jr.



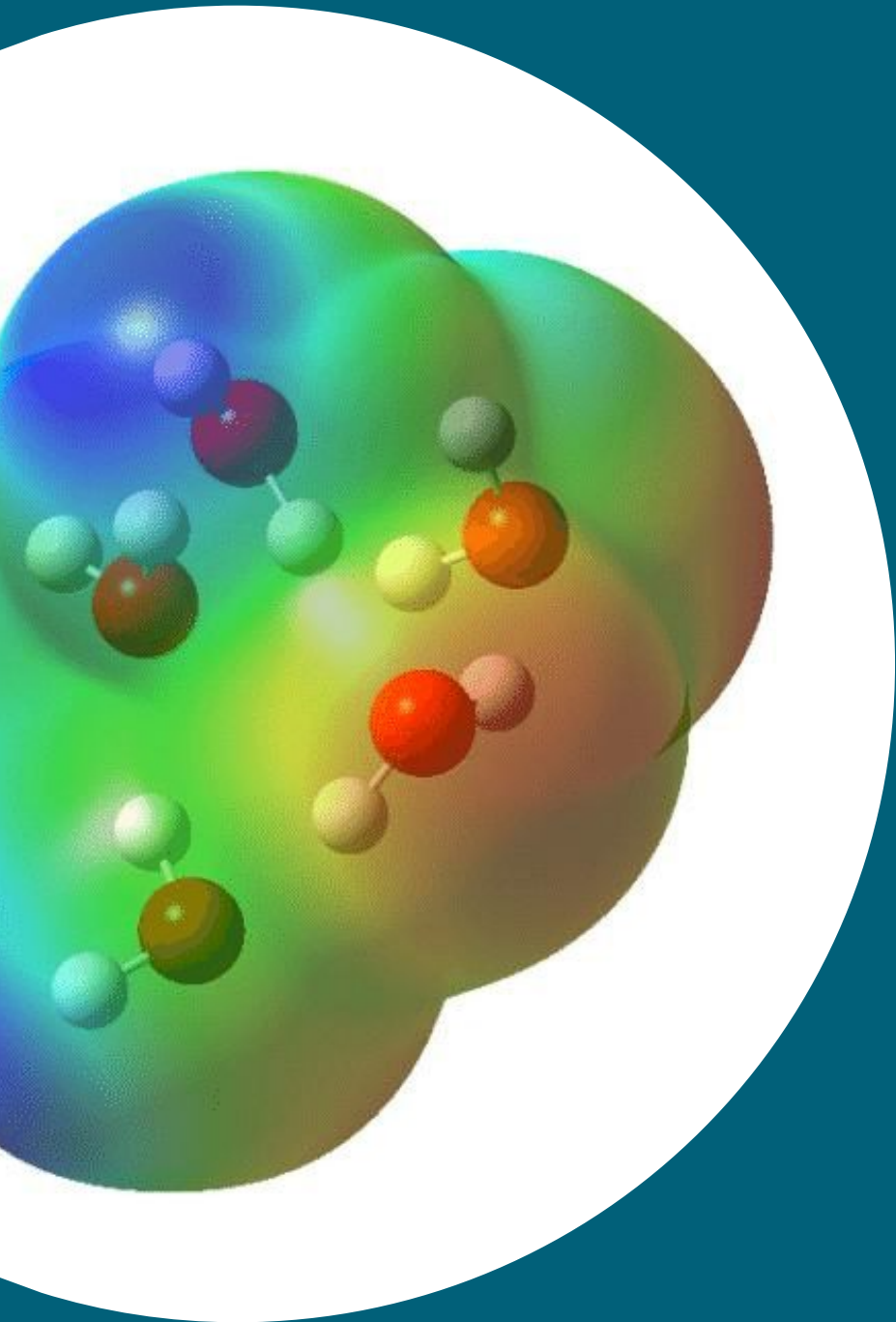
Prof. Carla Manske



Prof. Ingrid Oliveira

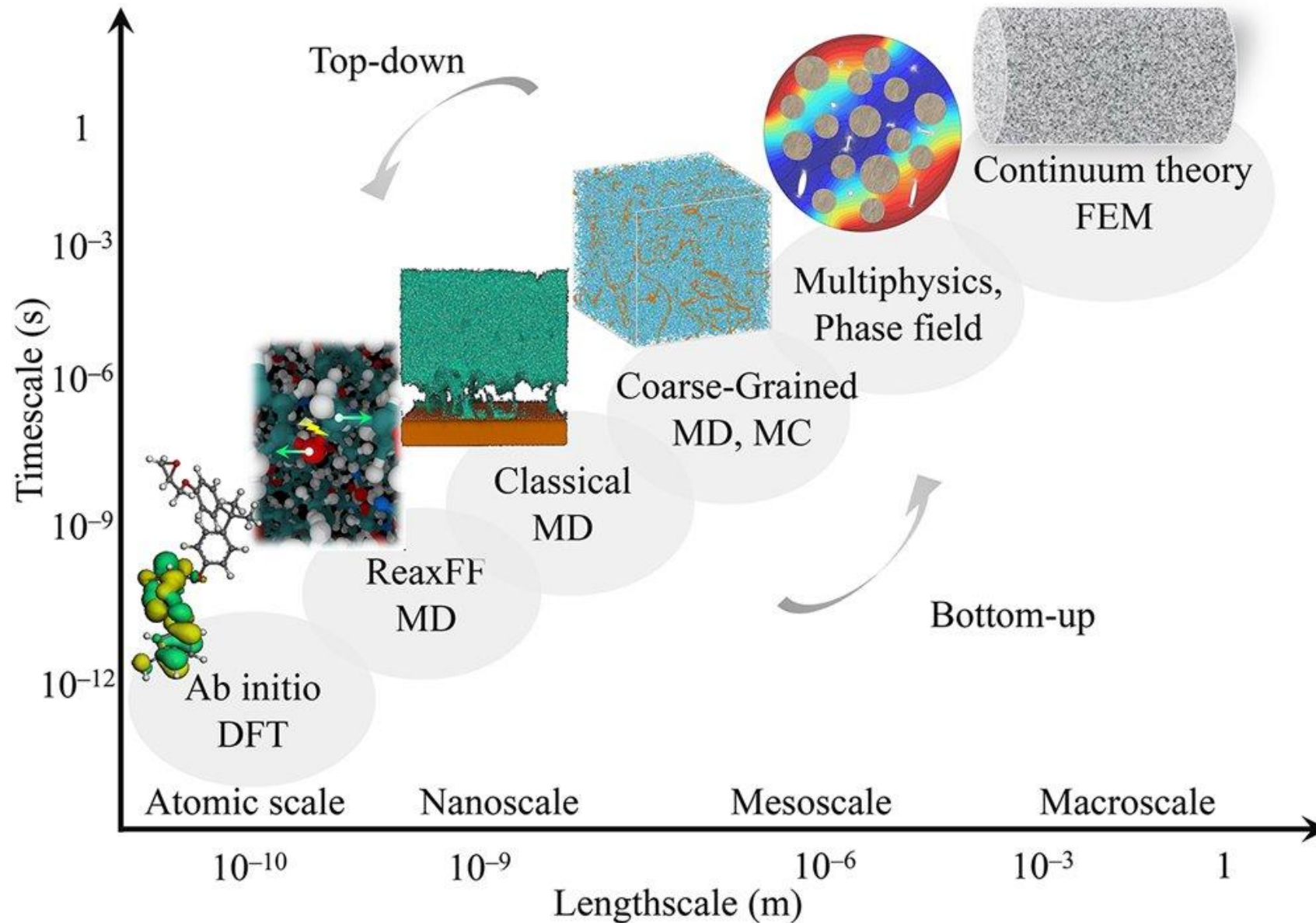


Prof. Elvis Soares



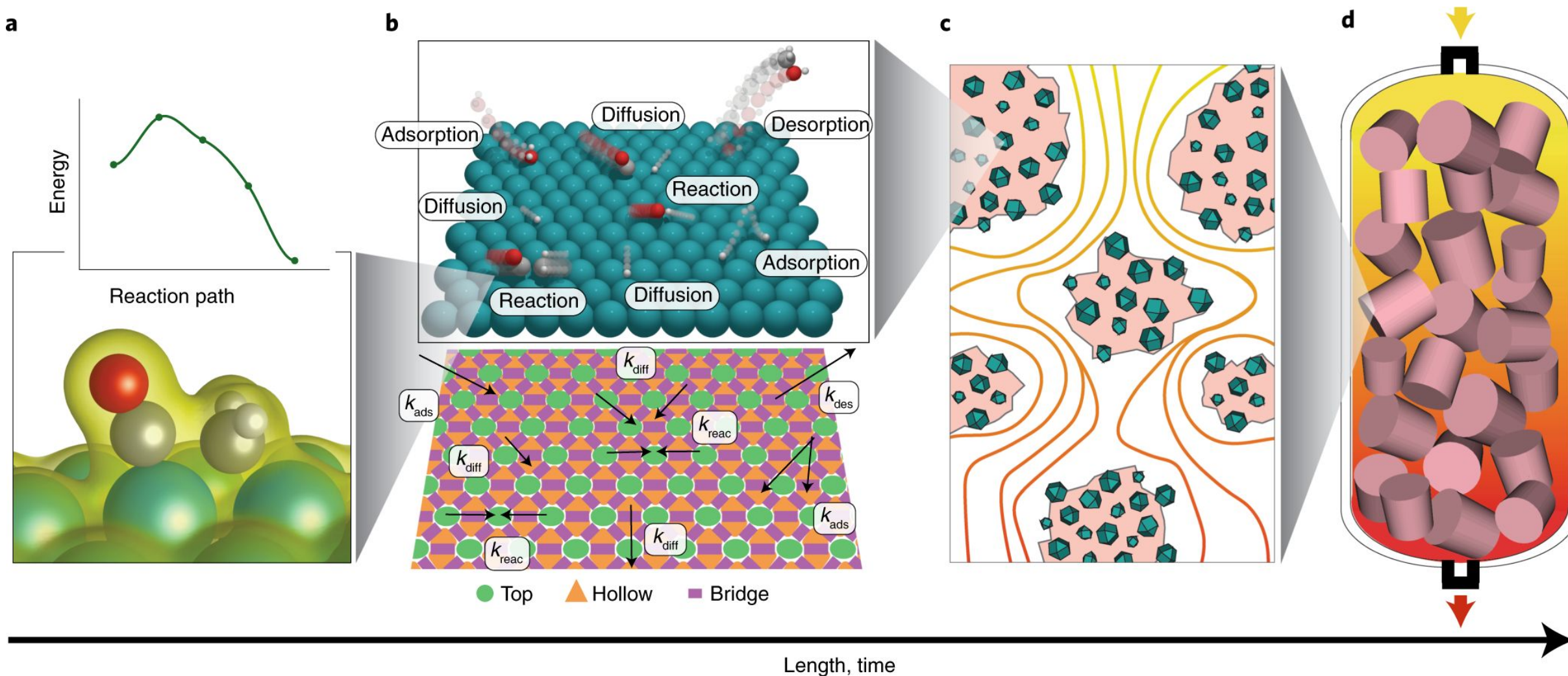
Áreas de Pesquisa

Modelagem Multi-Escala

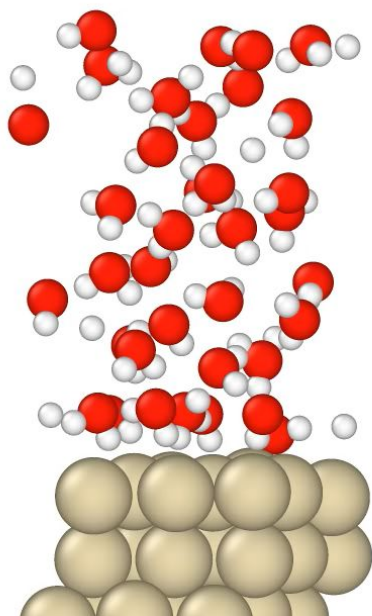


Sistemas Físico-Químicos Multi-Escala

Do Átomo ao Produto

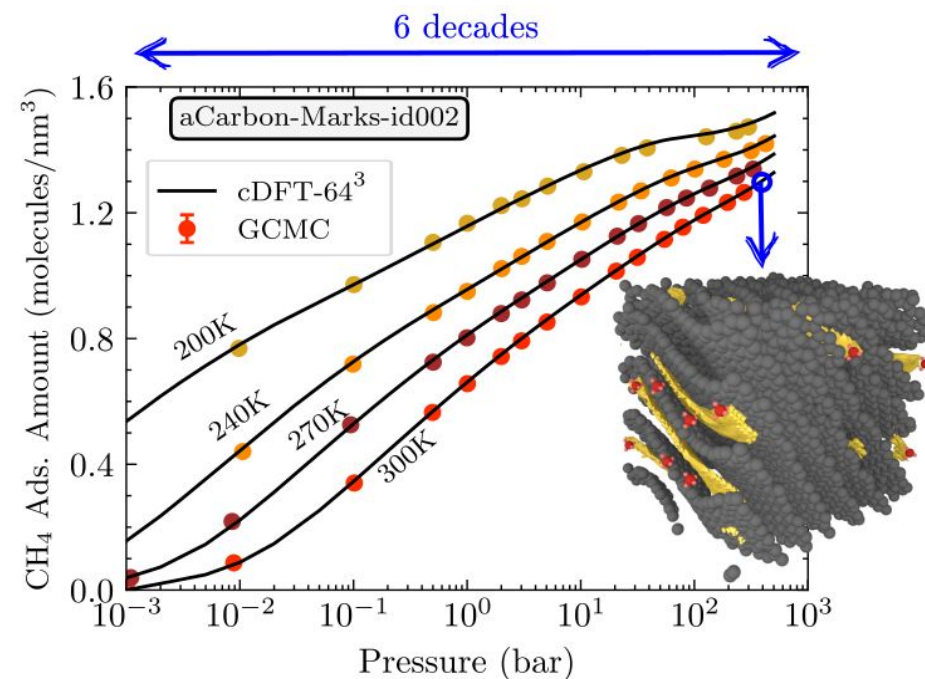
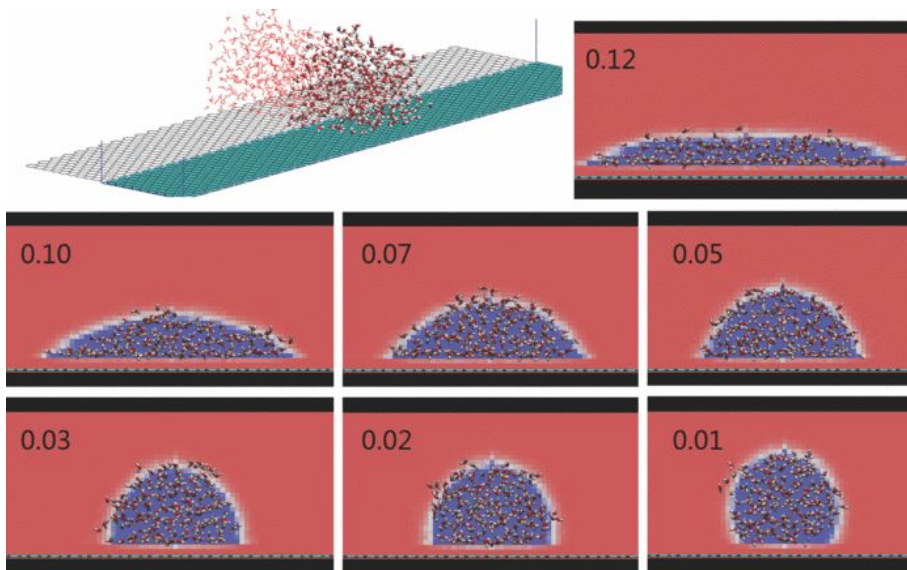


Cálculos Clássicos



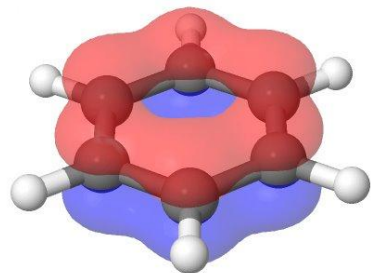
Dinâmica
Molecular

Fenômenos
Interfaciais

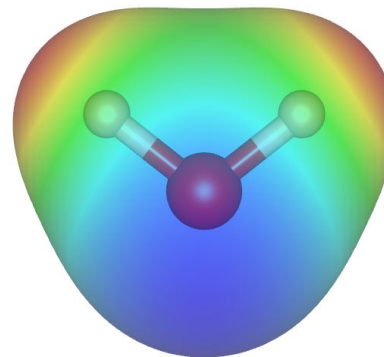


GCMC/cDFT

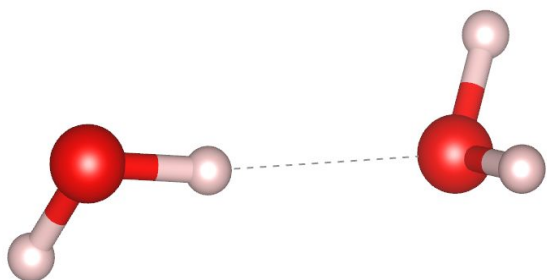
Cálculos Quânticos



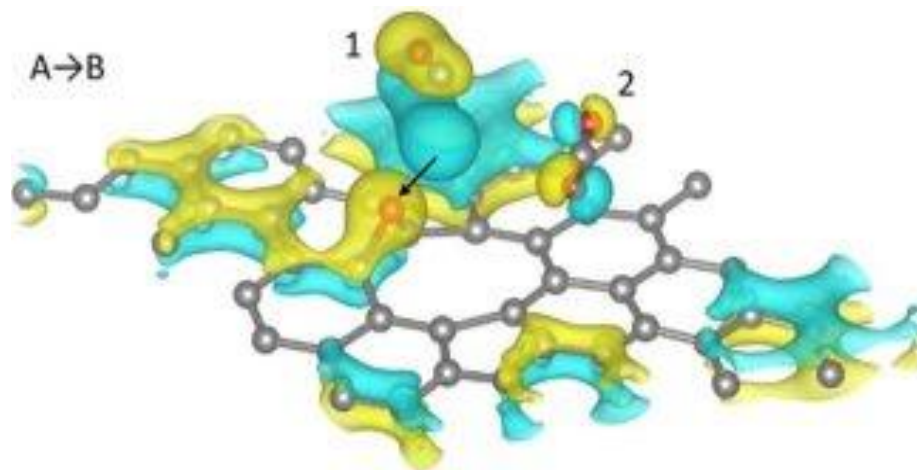
Configuração
Eletrônica



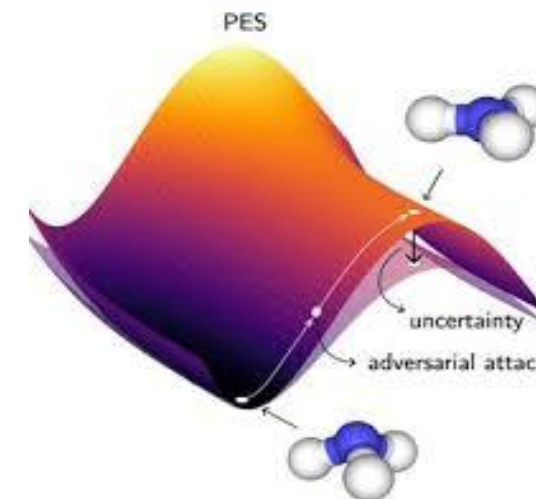
Análise Parcial de
Cargas



Interações
Intermoleculares

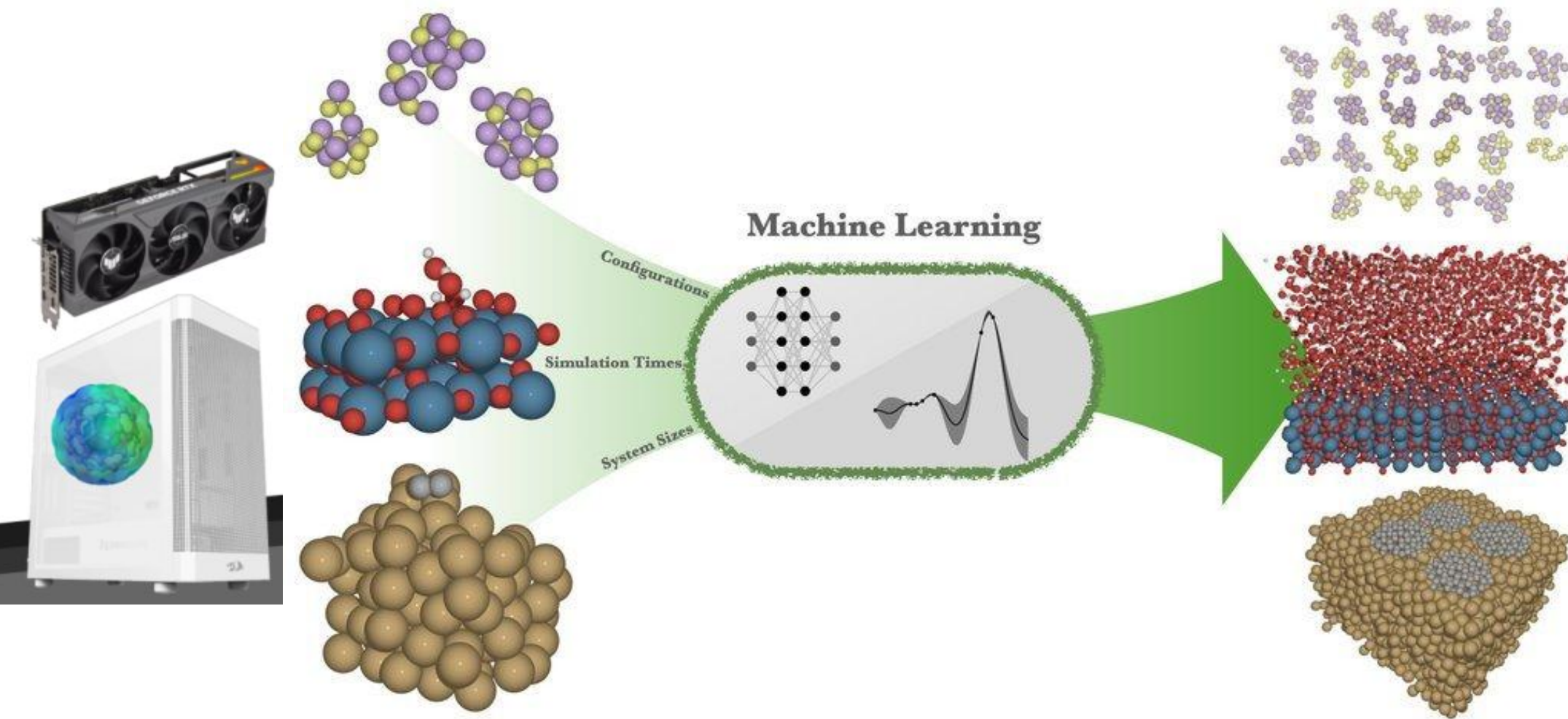


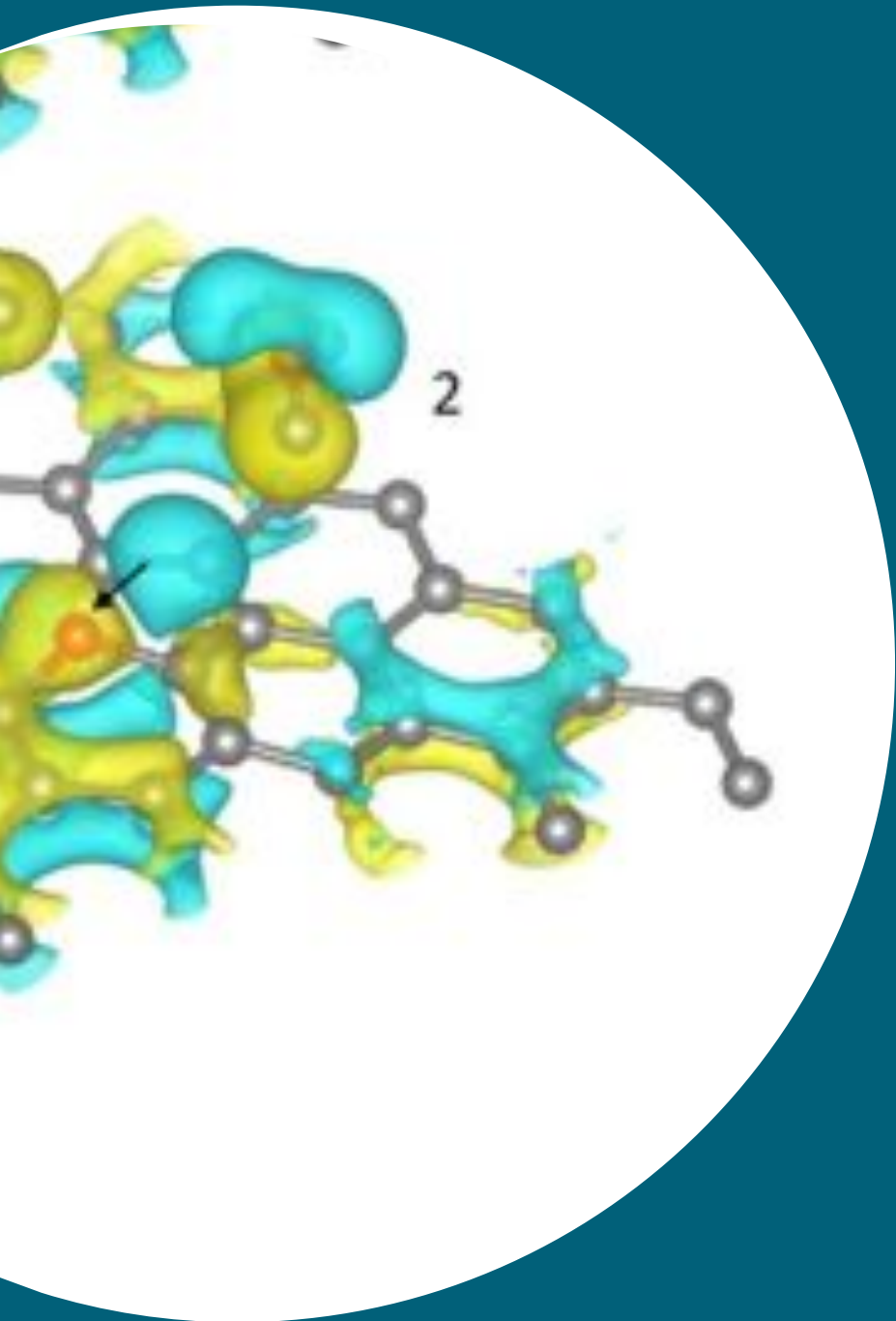
Ligações Químicas



Geração de Campo
de Força

IA científica





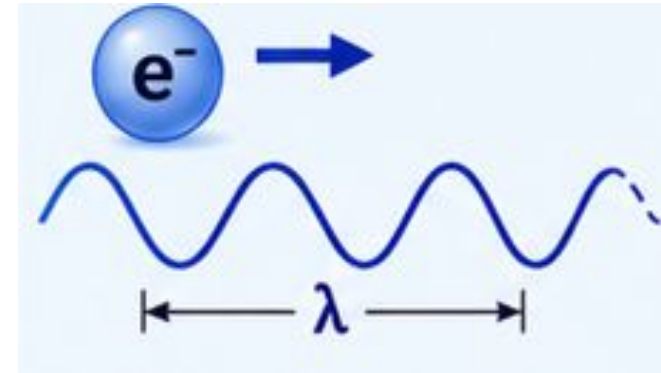
Cálculos Ab Initio

Teoria Quântica

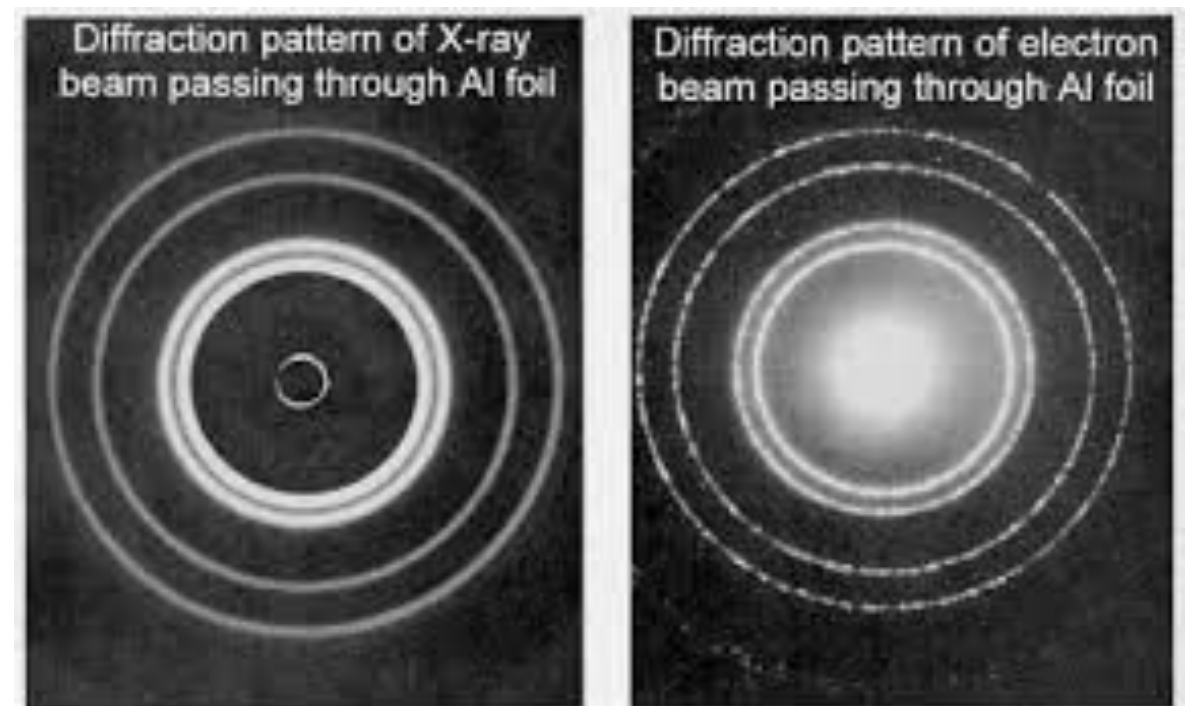
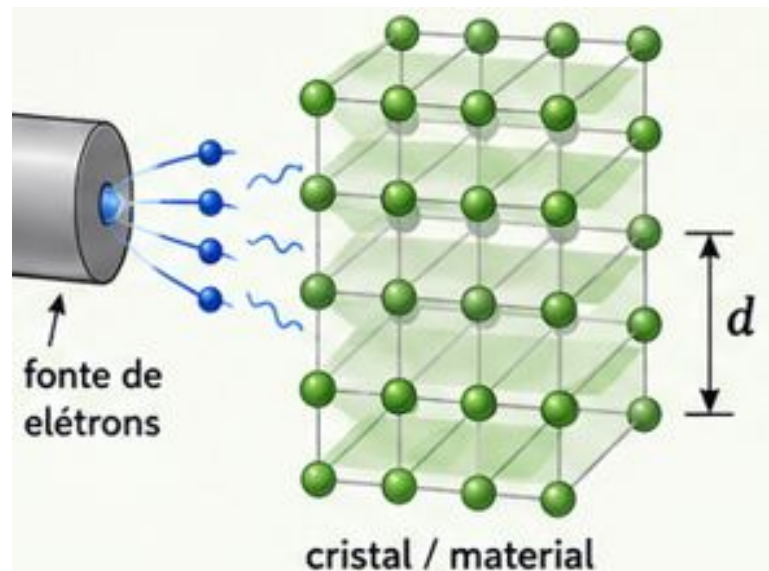
$$p = \frac{h}{\lambda}$$



de Broglie

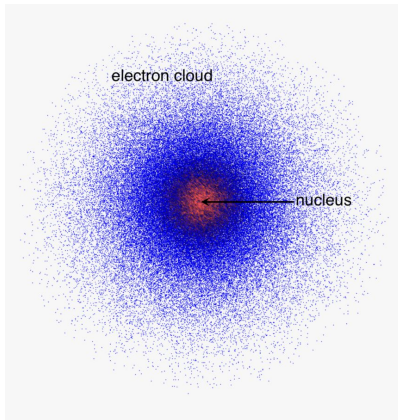


Elétrons se comportam como
partícula e como onda



Teoria Quântica

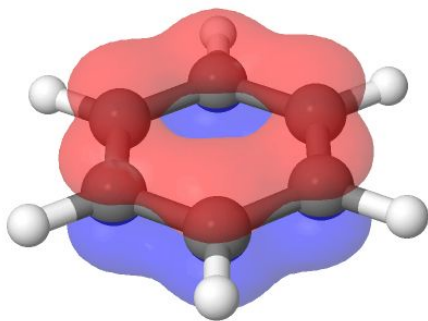
$$\hat{H}|\Psi\rangle = E|\Psi\rangle$$



Schrodinger

Teoria Quântica

$$\hat{H}|\Psi\rangle = E|\Psi\rangle$$



Schrodinger



Slater



Hartree



Fock

Problema de muitos corpos

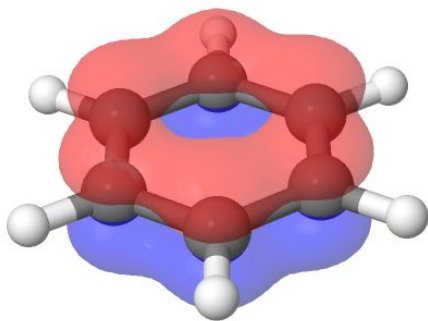
$$\hat{H}_e = -\frac{1}{2} \sum_{i=1}^N \nabla_i^2 - \sum_{A=1}^M \sum_{i=1}^N \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{2} \sum_{A=1}^M \sum_{B \neq A}^M \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}$$

**Energia do estado
fundamental**

$$E_0 = \frac{\langle \Psi_0 | \hat{H}_e | \Psi_0 \rangle}{\langle \Psi_0 | \Psi_0 \rangle}$$

Teoria Quântica

$$\hat{H}|\Psi\rangle = E|\Psi\rangle$$



Schrodinger



Slater



Hartree



Fock

Problema de muitos corpos

$$\hat{H}_e = -\frac{1}{2} \sum_{i=1}^N \nabla_i^2 - \sum_{A=1}^M \sum_{i=1}^N \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

**Energia do estado
fundamental**

$$E_0 = \frac{\langle \Psi_0 | \hat{H}_e | \Psi_0 \rangle}{\langle \Psi_0 | \Psi_0 \rangle}$$

Método “ouro”: CCSD(T)

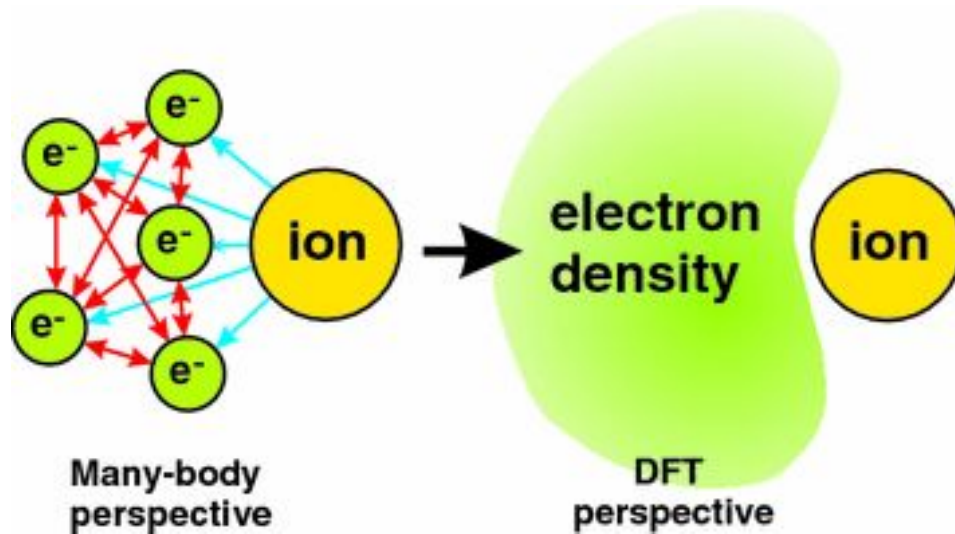
Métodos de perturbação: MP2

Dificuldades:

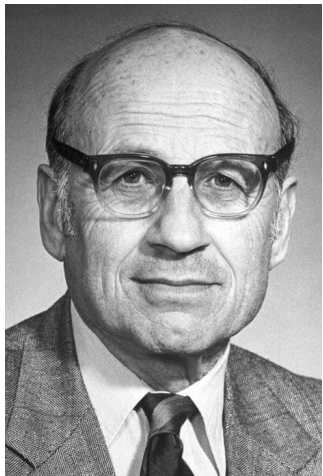
- Moléculas grandes
- Dificuldade de paralelização

Teoria do Funcional da Densidade – DFT

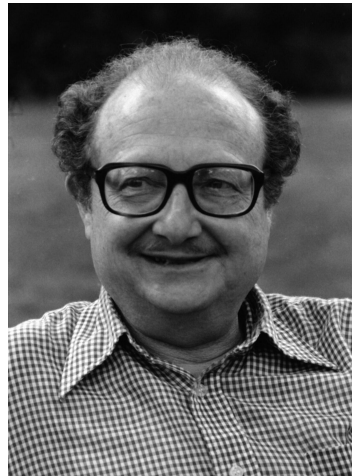
Energia como funcional da densidade



$$E[n(\mathbf{r})] = E_{HK}[n(\mathbf{r})] + \int n(\mathbf{r})V_{\text{ext}}(\mathbf{r})d\mathbf{r}$$



Kohn



Hohenberg



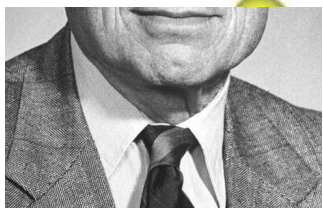
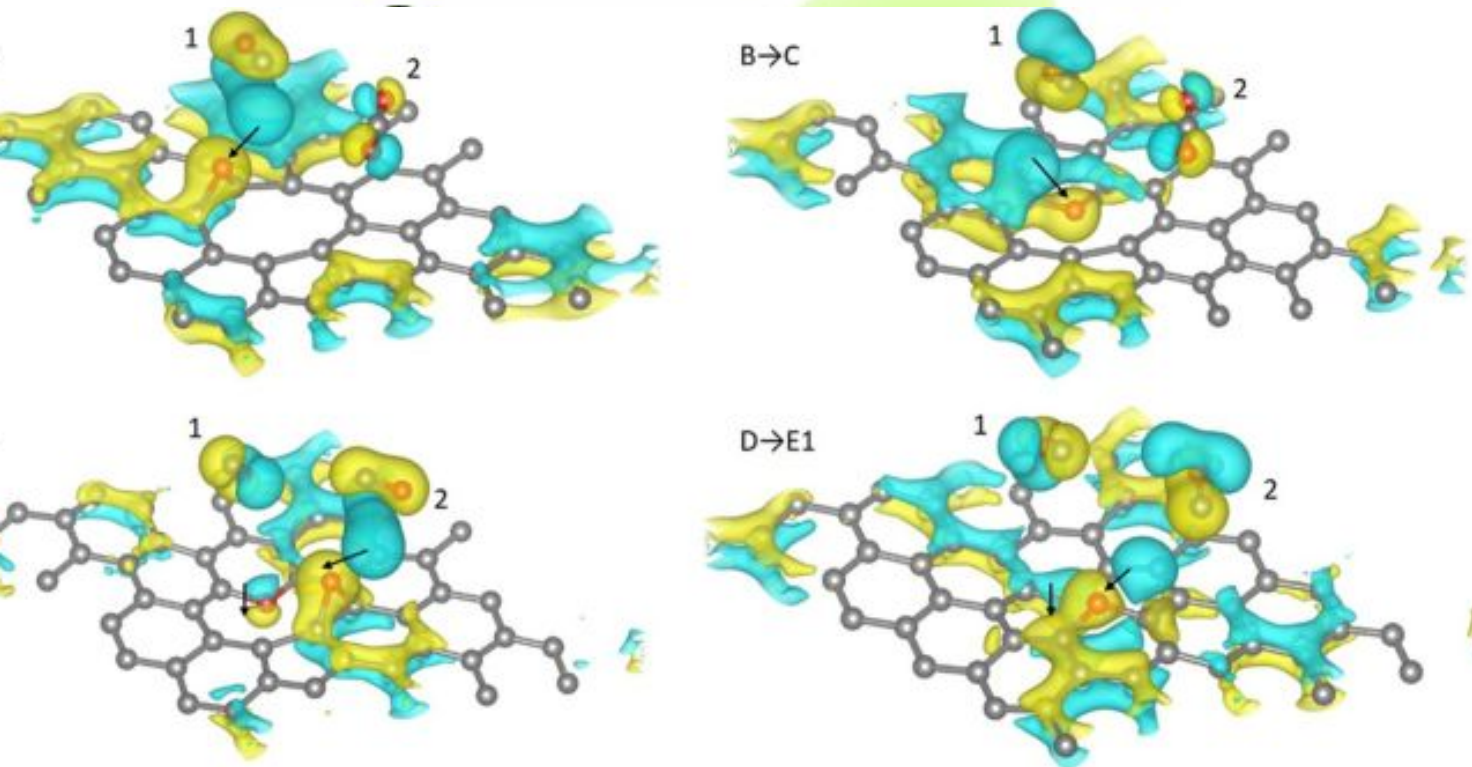
Sham

Energia do estado fundamental

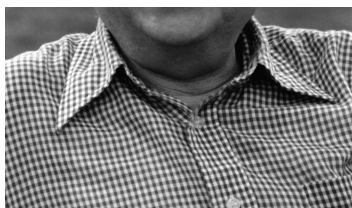
$$E_0 = E[n_0(\mathbf{r})]$$

Teoria do Funcional da Densidade – DFT

Energia como funcional da densidade



Kohn



Hohenberg



Sham

$$E[n(\mathbf{r})] = E_{HK}[n(\mathbf{r})] + \int n(\mathbf{r})V_{\text{ext}}(\mathbf{r})d\mathbf{r}$$

Ótimo para sistemas grandes

Trata moléculas e sólidos

Adsorção, reação, catálise,...

Dificuldades:

- Forma dos funcionais

Método de Kohn-Sham

Funcional de Hohenberg-Kohn

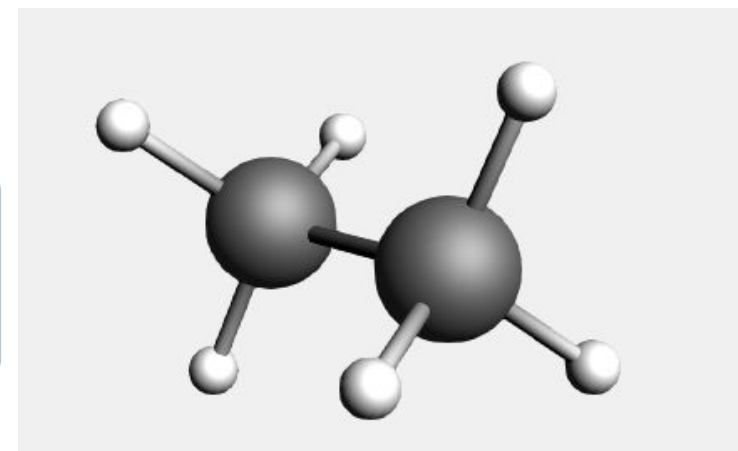
$$E_{HK}[\rho(\mathbf{r})] = \sum_i \langle \psi_i | -\frac{1}{2} \nabla_i^2 | \psi_i \rangle + \frac{1}{2} \iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' + E_{xc}[\rho(\mathbf{r})]$$

Densidade e orbitais de Kohn-Sham

$$\rho(\mathbf{r}) = \sum_{i=1}^N |\psi_i(\mathbf{r})|^2$$

Potencial Externo

$$V_{\text{ext}}(\mathbf{r}) = - \sum_{A=1}^M \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|}$$

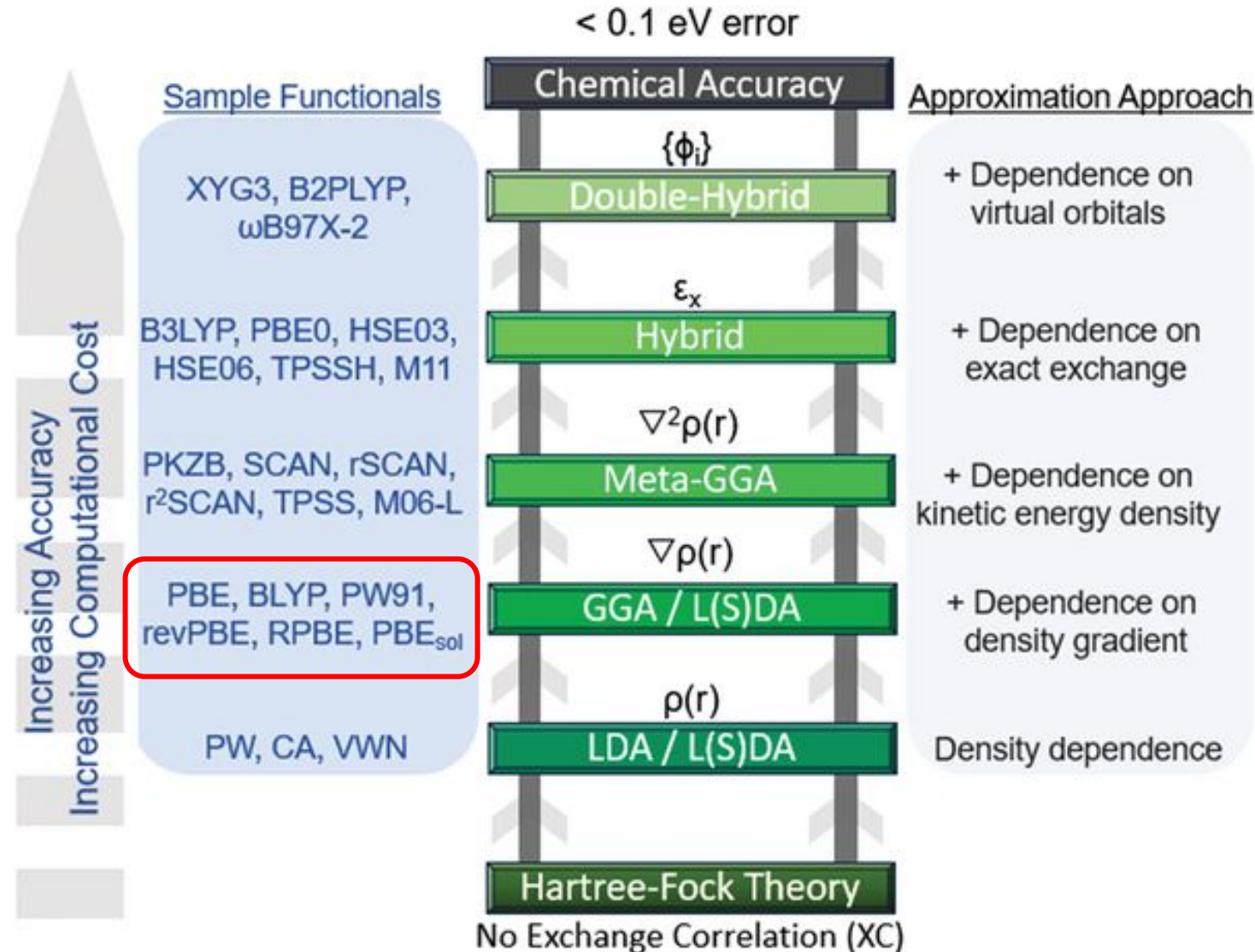


Equações de Kohn-Sham

$$\left[-\frac{1}{2} \nabla^2 + V_H(\mathbf{r}) + V_{XC}(\mathbf{r}) + V_{\text{ext}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r})$$

Funcionais de Troca e Correlação

$$E_{xc}[n(\mathbf{r})]$$



Conjunto de Funções de Base

Orbitais de Kohn-Sham

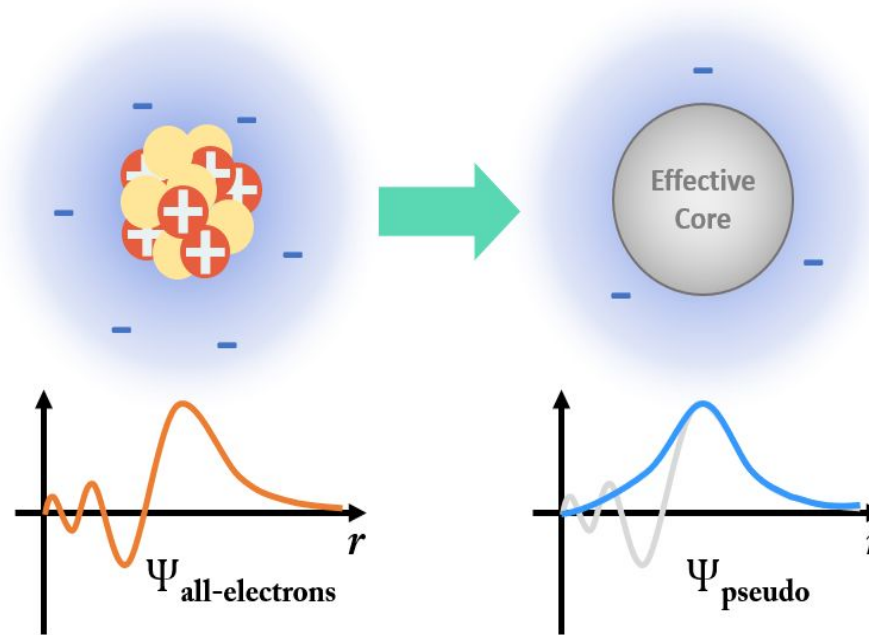
$$|\psi_i\rangle = \sum_{\mu=1}^B c_{\mu i} |\phi_{\mu}\rangle$$

- Base de ondas planas

$$|\phi_{\mu}\rangle = \sum_{i=1}^n d_{i\mu} |\mathbf{k}_i\rangle$$

- Bases gaussianas

$$|\phi_{\mu}\rangle = \sum_{i=1}^n d_{i\mu} |\text{GTO}_i\rangle$$

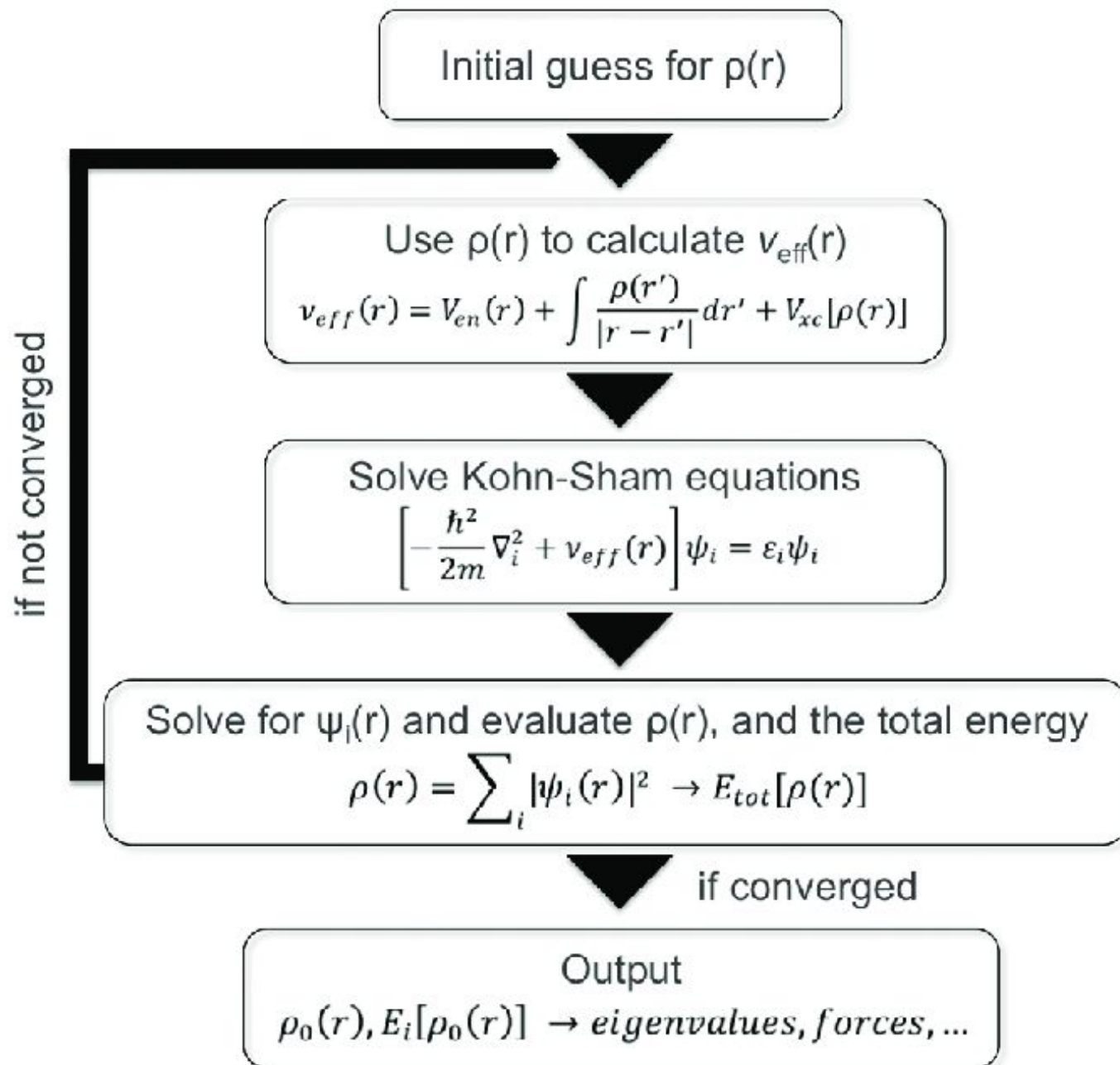


The number of gaussian functions summed to describe the inner shell orbital

The number of gaussian functions summed in the second STO

3-21G

The number of gaussian functions that comprise the first STO of the double zeta

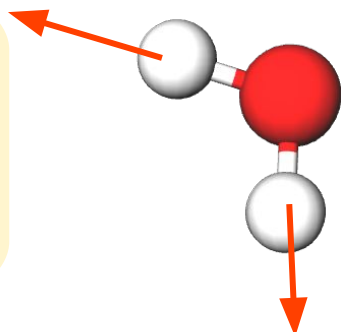


Otimização de Geometria Molecular

$$\frac{\partial E}{\partial \mathbf{R}_A} = \frac{\partial}{\partial \mathbf{R}_A} \left[- \sum_{A=1}^M \int \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \rho(\mathbf{r}) d\mathbf{r} + \frac{1}{2} \sum_{A=1}^M \sum_{B \neq A}^M \frac{Z_A Z_B}{|\mathbf{R}_B - \mathbf{R}_A|} \right]$$

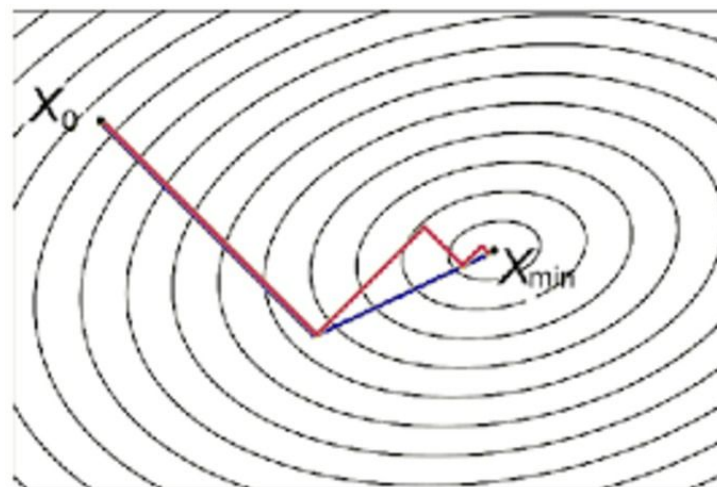
Força no átomo A

$$\mathbf{F}_A = - \frac{\partial E}{\partial \mathbf{R}_A}$$



Cálculo do Equilíbrio das posições

$$\mathbf{R}_A^{n+1} = \mathbf{R}_A^n + \alpha \mathbf{F}_A^n$$



Diferentes Métodos:

- Método de Newton
- BFGS
- FIRE

Ferramentas Computacionais

Bases Gaussianas



GÅMESS

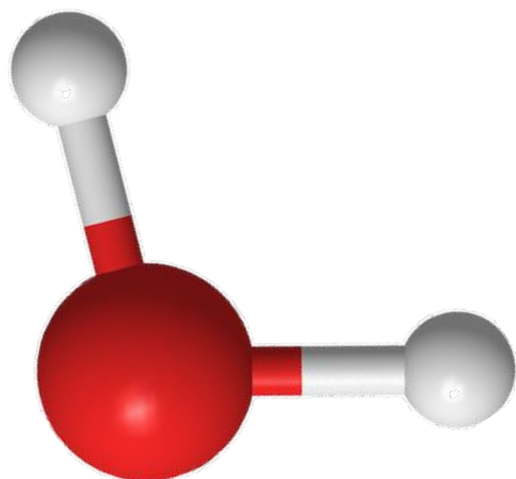


NWCHEM

Bases de Ondas Planas

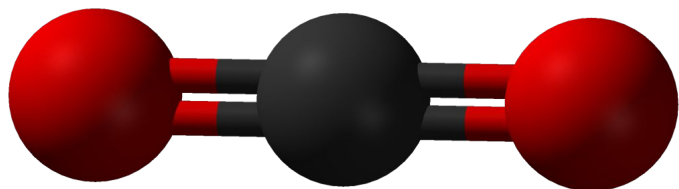


Aplicação para H2O



Nome	Método	ligação (Å)	ângulo (°)
	LDA/sto-3g		
	LDA/6-31++G**		
	LDA/cc-pvtz		
	PBE/sto-3g		
	PBE/6-31++G**		
	PBE/cc-pvtz		
	B3LYP/sto-3g		
	B3LYP/6-31++G**		
	B3LYP/cc-pvtz		
	Experimental	0.958	104.48

Aplicação para CO₂



Nome	Método	ligação (Å)	ângulo (°)
	LDA/sto-3g		
	LDA/6-31++G**		
	LDA/cc-pvtz		
	PBE/sto-3g		
	PBE/6-31++G**		
	PBE/cc-pvtz		
	B3LYP/sto-3g		
	B3LYP/6-31++G**		
	B3LYP/cc-pvtz		
	Experimental	1.17	180

OBRIGADO!



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